

MARSHALL WACE

Technology Early Careers Application Guide

Graduate, Early Careers & Intern Programmes

We know many students find interviewing to be stressful and overwhelming, which is why we want to make the process as transparent as possible. This allows you to properly prepare and give each stage your best.

Our aim is to create an engaging process where you are challenged so we can evaluate your skills. We are very intentional with who we hire and seek out the top candidates to ensure we are best able to continue the best service to our clients.

In this document you will find additional information on the different stages of the interview process, tips on how to best prepare, and the skills we are looking for. This guide covers both our Graduate and Intern programmes. The interview process is the same for both, with one additional stage for Graduate candidates which is noted where applicable.

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Marshall Wace is an equal opportunity employer. Individuals seeking employment are considered without regard to race, colour, religion, national origin, age, sex, marital status, ancestry, physical or mental disability, veteran status, sexual orientation, or any other category protected by applicable law.



Application

You need to fill out our application form online, making sure to submit your most recent CV.

Your CV should include key details like:

- Which university you are attending
- Which course you are studying
- Your start and end dates
- Details of any previous work experience
- Any notable technology projects you have worked on – particularly any personal or non-university projects

There will also be a series of questions to answer, including a written response to the following: Outside of AI and ML, what's a technology, tool or project that has caught your attention recently? What made you want to learn more about it, and have you tried it out?

This will be reviewed by a member of our Human Capital team and may come up in your screening interview, so we recommend choosing something you could discuss in some depth and being aware of your audience in your write-up. We'd love to read this in your own words – we're interested in what genuinely excites you, not a polished essay.

You do not need to submit a cover letter, but you are welcome to if you feel it supports your application.





Interview Stages

Step One: Codility Assessment

We will send you an online programming assessment via Codility. This test will investigate your general coding aptitude, error handling and sequence processing. Some questions will allow you to use the language of your choice, others may require you to use a specific language you may not be familiar with – this is by design.

Your invite will be individual and sent directly by Codility. You can begin the Codility assessment at any time of your choosing but it should be completed within 7 days. Please check junk mail folders if you do not see the Codility email in your inbox. If you still cannot find it, please contact your recruiter.

Please note, our tests are private and confidential. By participating in our test, you agree that you will hold the information as private and confidential and will not share the details outside of our organisation. This is important to ensure equity in our process and we thank you in advance for your compliance.

If you did not inform us that you required adjustments in your application but do require them, please email us before starting your test. Once you have opened the test and seen the questions, we will not be able to send you an adjusted version. This is to ensure fairness across candidates.

Preparation Tips:

- There are many online resources that will help you to prepare for an online test of this nature, including some created by Codility itself, found on their website.

Step Two: Screening Call



If you perform well in the Codility assessment, you will be invited to a 15-30-minute screening with one of our recruitment team via Zoom.

They will be looking to understand the following:

- Your interest in the programme
- Your interest in technology
- Your interest in Marshall Wace
- Your core technical and soft skills

To prepare for this call, you should consider the above and make sure you are in a quiet place with suitable access (e.g. a stable internet connection, camera etc). If you do not have access to suitable technology, let us know and we can see what adaptations can be made.

Step Three: Online Lab

The Online Lab is a practical assessment designed to evaluate your technical skills, problem-solving ability and communication in a realistic working environment. Reasoning about code, debugging code – language agnostic but examples in python.

The assessment has two parts:

1. Solo Working Session (~60 minutes)
You'll log into a self-contained virtual desktop and work through a series of questions and tasks independently. These vary in type and complexity — some are shorter and more focused, others require deeper exploration and investigation.

You'll have access to a web browser and other tools within the environment, and you're welcome to use them however you find helpful. However, the environment is locked down — you must work entirely within the virtual machine and are not permitted to use any resources outside of it.

Before your session, you'll need to download and install **VMware Horizon Client** to connect. Instructions and access details will be sent to you in advance.

2. Debrief Call (~20–30 minutes)
After the solo session, you'll join a video call with one or two members of our technology team to walk through your work. This is a conversation, not a test — they'll ask about your approach, your reasoning, and how you worked through the problems.

The debrief is an important part of the assessment. We're just as interested in *how* you reached your answers as the answers themselves.

What We Are Assessing

We want to understand how you think through technical problems, how clearly you communicate your reasoning, and how you handle working independently. Don't worry about getting everything perfect — we'd rather see a well-reasoned approach to a problem than a rushed answer.

Preparation Tips:

- **Get your setup ready early.** Make sure you have a stable internet connection and are comfortable with the laptop or PC you'll be using. Download and install VMware Horizon Client before your session — don't leave this to the last minute.
- **Read the instruction guide carefully.** It will be sent with your invitation and contains everything you need to know about accessing the environment.
- **Brush up on reading code.** You don't need to be an expert in any particular language, but being comfortable with common programming concepts (variables, loops, functions, data structures) will help.
- **Practice thinking out loud.** In the debrief, we'll ask you to explain your reasoning. Practising how you'd walk someone through a problem — what you tried, what you ruled out, and why — will help you feel more comfortable.
- **Don't panic if something is unfamiliar.** You will likely encounter things you haven't seen before. That's deliberate — part of what we're assessing is how you approach the unknown. Stay calm, read carefully, and work through it methodically.





Step Four: Virtual Technical Interview

This is a one-hour virtual interview conducted over Zoom with two members of the Technology team. You will be asked one or two technical questions that explore your knowledge of key programming and computer science concepts. The questions will typically develop in complexity over the course of the interview and could cover areas such as data structures, algorithms, object-oriented design, systems architecture, data processing or complexity analysis.

We do not typically use a live-programming environment. Instead, we are looking to understand how you think through a problem. We will ask about your approach, your solution's efficiency, potential improvements and alternative approaches. We may also ask about projects or skills on your CV, focusing on your specific contributions and how you applied technical concepts in practice.

Preparation Tips:

- Review the fundamentals of data structures, algorithms, object-oriented principles and common design patterns.
- Practice explaining your thinking out loud. Being able to talk through your reasoning clearly is as important as arriving at the right answer. Even if you are unsure, structuring your thought process helps us understand how you approach problems.
- Ask clarifying questions. Taking a moment to understand the problem fully before diving in is a strength, not a weakness.
- Think about trade-offs. For many questions, there is no single right answer. Be prepared to discuss the advantages and disadvantages of different approaches.
- Be ready to discuss the technical detail of any projects listed on your CV.
- Set up your environment (camera, microphone) in a quiet, well-lit space. Have pen and paper ready to sketch out ideas if needed.

Step Five: On-site Interviews

The on-site stage consists of multiple interviews conducted at one of our offices. These must take place in person, though not necessarily at the office you are applying to. We will work with you to find the most convenient location for you and our interviewers.

These are knockout rounds, which means we may conclude the process early if we do not feel that an earlier stage has gone well. This is to save both the interviewers' and your time.

Graduate candidates will have up to five interviews. Intern candidates will have up to four (the Senior Leadership interview applies to graduates only).

You will also typically have a coffee break or lunch with a recent graduate, depending on the location and timing of your interview. This is to give you a chance to find out more about what it is like to join us.

a) Project Presentation (45 minutes)

We ask that you prepare a presentation about code you have previously written. You will present to Marshall Wace technology team members, who will ask probing questions about your project and code to assess your understanding and passion.

As this takes place on-site, you will need to bring your own laptop to demo your project. If you do not have one available, we can provide a laptop – in that case, please ensure your code is accessible on GitHub (or a similar platform) beforehand.

You are welcome to present however you prefer, but we must see actual code. Your introduction should be long enough to explain your project clearly (10–15 minutes usually suffices) but should leave plenty of time for the interviewers to read your code and discuss it with you.

Project Guidance:

What you choose to present is up to you. Examples include a website or game you have built, a personal project, or a script you have written to solve a problem. The aim is to understand your programming ability, technical depth, passion for technology, and what skills you would bring to Marshall Wace.

We recommend that you:

- **Choose a project with genuine technical depth.** Avoid tutorials you have simply followed or basic implementations. Our most successful candidates present projects that demonstrate substantial technical challenges they have overcome.
- **Choose a project that distinguishes you.** We interview students from top universities internationally who share similar coursework, so we frequently see the same group projects. A personal or self-directed project will stand out far more than a standard assignment.
- **Know your code inside and out.** You must be able to explain every technical decision, including architecture choices, technology selection and implementation details. If it has been a while, re-read your code before the interview.
- **Set the scene before showing code.** Start with the problem you were solving, why it mattered to you, and a brief overview of your approach. This helps your interviewers follow along and shows you can communicate the big picture.
- **Be ready to justify your decisions.** We will ask why you chose specific technologies, architectures or approaches. Think about what alternatives existed and why you went the direction you did.
- **Show awareness of code quality.** Be prepared to discuss testing strategies, error handling, security considerations and design patterns.

For group projects, we will look to understand your individual contributions and expect a good understanding of your partners' work.

Preparation Tips:

- A few slides to structure your presentation can help, but are not required. If using slides, engage your interviewer rather than reading from them.
- Have your code ready to show and be prepared to walk through any part of it in isolation.
- Be aware that research-heavy projects may not give us the best understanding of your programming ability.
- Ensure your laptop is charged and ready. If using a provided laptop, make sure your code is on GitHub beforehand.

The most successful candidates demonstrate genuine technical depth, clear ownership of their work, and excellent communication skills. They respond thoughtfully to questions, showing both technical competence and intellectual curiosity.

b) Technical Interview (45 minutes)

This interview is led by two members of our Technology team. You will typically be asked one or two programming questions to solve while explaining your working. We are looking to understand how you think and approach problems, as well as your ability to answer the questions themselves.

Depending on time, you may also be asked about details on your CV – for example, projects you have worked on, your degree course, or what excites you about technology.

c) Manager / Cultural Interview (1 hour)

This interview will be conducted by two senior members of our Technology team. They will ask competency-based questions to gain an understanding of your behaviours and cultural fit at Marshall Wace. They may also probe technical areas if earlier interviewers identified gaps worth exploring further.

d) Human Capital Interview (30 minutes)

This interview will be taken by one of our Human Capital team. They will be looking to gain an understanding of your background, your interest in the programme and the business, and will answer any remaining questions you have.

e) Senior Leadership Interview (Graduate Programme Only)

This stage applies to Graduate candidates only and does not form part of the Intern interview process.

If you perform well in all of the above stages, you will meet one of our senior technology leaders. This could be the CTO, the Global Head of Human Capital, the Chief of Staff for Technology, or a Senior Technology Manager. This interview will focus on your motivations, your passion for technology, and your fit for the programme.

On-site Preparation Tips:

- Be prepared to explain how you reach your technical solutions. Think out loud – we want to see your reasoning, not just your answer.
- Do not be afraid to ask questions or re-clarify anything you did not understand.
- Use specific examples to back up your answers during competency-based questioning. The STAR method (Situation, Task, Action, Result) is a useful framework.
- Practice explaining technical concepts simply. Several of your interviews will be with people who want to see that you can make complex ideas accessible.
- Be prepared to reflect on past experiences – what went well, what you would change, and what you learned.
- Have opinions. We want to understand how you think, so do not be afraid to share your perspective, even if you are not certain.
- Have some questions prepared for your interviewers about things you would like to find out more about.



Feedback

We will always aim to give you feedback within one working week of your interview or test via email.

If you have any questions or would like more information, please get in touch with your recruiter and they will try to reply as soon as possible. If there are any delays, we will do our best to keep you informed. Please be aware it is a small team managing tens of thousands of applications so emails may not be responded to immediately.

FAQs

I have been rejected this season. Can I reapply?

No, in the interest of fairness across candidates we will not accept requests to reapply in the same recruitment season (i.e., for roles starting in 2027). This includes if you did not pass the Codility test or screening call.

I was rejected last season. Can I reapply?

Yes, we welcome re-applications from those who did not make it through the process last year. However, being progressed in previous years does not guarantee that you will be in the current year.

I have been in process before. Can I skip any stages?

We will review these requests on a case-by-case basis depending on the feedback available and which stages you previously completed. Where possible, we like everyone to complete all the stages to ensure fairness across candidates.

Can I interview on weekends or evenings?

We will not arrange weekend interviews. Our employees' wellbeing is important to us, and we do not expect them to be conducting interviews outside of their working hours. Typically, we will try to schedule virtual interviews between 8am and 6pm GMT as many of our interviewers are based in the UK. For on-site interviews, we will aim to conduct these between 8:30am and 5pm local time.

Can I apply for multiple offices?

We ask that you only apply to one office. The process we follow and the quality we look for is the same across all offices, so applying to multiple locations will not benefit your application but will delay its review.

Can I arrange a call or coffee to discuss my application before submitting it?

No. We receive a very high number of applications for each international office and are a lean team. If we conducted calls with every candidate ahead of starting their process, we would have no time to process the applications.

Can my travel be expensed?

We will reimburse reasonable travel for on-site interviews. To do so, we will request you complete a form that will be sent to you and attach receipts. We will not be able to reimburse without receipts.

What should I wear to interview?

We recommend business-smart attire. Many employees wear shirts or blouses with trousers, skirts or dresses.